Nithiyanandam S

Software Engineering Intern | HTML, CSS, JavaScript, React, Next.js, Python, C/C++

Chennai, Tamil Nadu, India | +91-7826060700 | nikeanand97@gmail.com | linkedin.com/in/nithiyanandam-s | github.com/Nike-Anand

Professional Summary

Software engineering student experienced in building responsive, user-friendly web applications for a 9-division hospital organization. Strong in OOP, C/C++, and Agile. Two-time Smart India Hackathon national finalist; published researcher.

Education

Sri Venkateswara College of Engineering 2023 – Expected May 2027

Bachelor of Engineering (B.E.) in Computer Science and Engineering | CGPA: 7.8/10 *Chennai, Tamil Nadu, India*

Technical Skills

Languages: HTML5, CSS3, JavaScript, TypeScript, Python, C, C++, Java, SQL
Frontend: React, Next.js, Tailwind CSS, Payload CMS, Responsive Design, Web UI Development, Accessibility (WCAG, ARIA)
Backend & Databases: Node.js, Flask, REST APIs, Microservices, MySQL, PostgreSQL, MongoDB, PL/SQL
AI / ML: PyTorch, OpenCV, OCR, Deep Learning, Natural Language Processing (NLP), Generative AI, Prompt Engineering
Tools & DevOps: Git, GitHub, Docker, Postman, Agile, SDLC, Microsoft Excel, Word, PowerPoint
Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Design, Cryptography
Soft Skills: Collaboration, Teamwork, Communication, Presentations, Project Management, Attention to Detail, Prioritization

Work Experience

Software Development Engineer Intern Mar 2026 – Aug 2026

Tech Machers, Chennai, Tamil Nadu, India

- Migrated websites of a **9-division** hospital organization from static pages to **Payload CMS**, improving **page load time** and **editing time**.
- Built reusable CMS blocks and a user-friendly admin panel with **HTML, CSS, JavaScript, React, Next.js, and Node.js**, letting non-technical staff update content independently; deployed with **Docker**.
- Tested the admin panel with real editors; applied **user-centered design** feedback to improve **usability**.
- Resolved **150+ P1–P3 bugs** across routing, rendering, and REST API integration through team collaboration, Git, and Agile methodology.

Data Analyst Intern Dec 2025 – Jan 2026

Vi (Vodafone Idea), Chennai, Tamil Nadu, India

- Analyzed telecom datasets spanning **millions of records** using **SQL and Python** to identify customer churn patterns and actionable retention insights.

AI/ML Intern & Generative AI Trainer Oct 2024 – Nov 2025

Altruisty, Chennai, Tamil Nadu, India

- Trained and mentored **20+ interns** via presentations on Generative AI, prompt engineering, and transformers.
- Built a product description-generator from scratch using **deep learning and NLP** on the **Amazon Product Dataset (approx. 48 million products)**, achieving **96% accuracy**.

Projects

Tamil Historical Character Recognition System | *Python, PyTorch, OpenCV, OCR* | GitHub

- Built an offline OCR pipeline for historical Tamil documents using a **CRNN (CNN + BiLSTM + CTC)** architecture, achieving **95%+ Tamil** and **98%+ English** character accuracy.
- Implemented adaptive thresholding, noise reduction, skew correction, and line/word segmentation for degraded manuscripts.

Lightweight Post-Quantum Cryptography for IoT | *Python, Flask, ESP8266, PQC* | GitHub | IJCRT Paper

- Designed a quantum-resistant encryption framework combining **lattice-based cryptography** with code-based error correction for resource-constrained IoT devices.
- Built a file encryption web app; led team to **Smart India Hackathon National Finalist**; published in **IJCRT (2025)**.

Bus Route Finder (Chennai MTC) | *Python, PyTorch, Graph Algorithms, HTML/CSS* | GitHub

- Scraped MTC route data into CSV/JSON and built a Python web app to compute optimal transit routes between stops.
- Applied **graph search algorithms** for route optimization; trained a compact language model for route queries.

Achievements & Certifications

- **Hackathons:** Smart India Hackathon National Finalist (**2024 & 2025**) | Infosys Hackathon 2025 (**Top 6 Nationwide**) | D3 Fest, IIIT Bhubaneswar (**State Winner, AIR 9**).
- **Programs, Research & Certifications:** Amazon ML Summer School 2026 Participant | First-Author PQC Paper (**IJCRT 2025**) | Google Cloud Generative AI Leader | NPTEL Computer Vision & NLP (**Elite + Gold**) | Anthropic AI Fluency & Claude 101 | Postman API Student Expert.